
MA1513

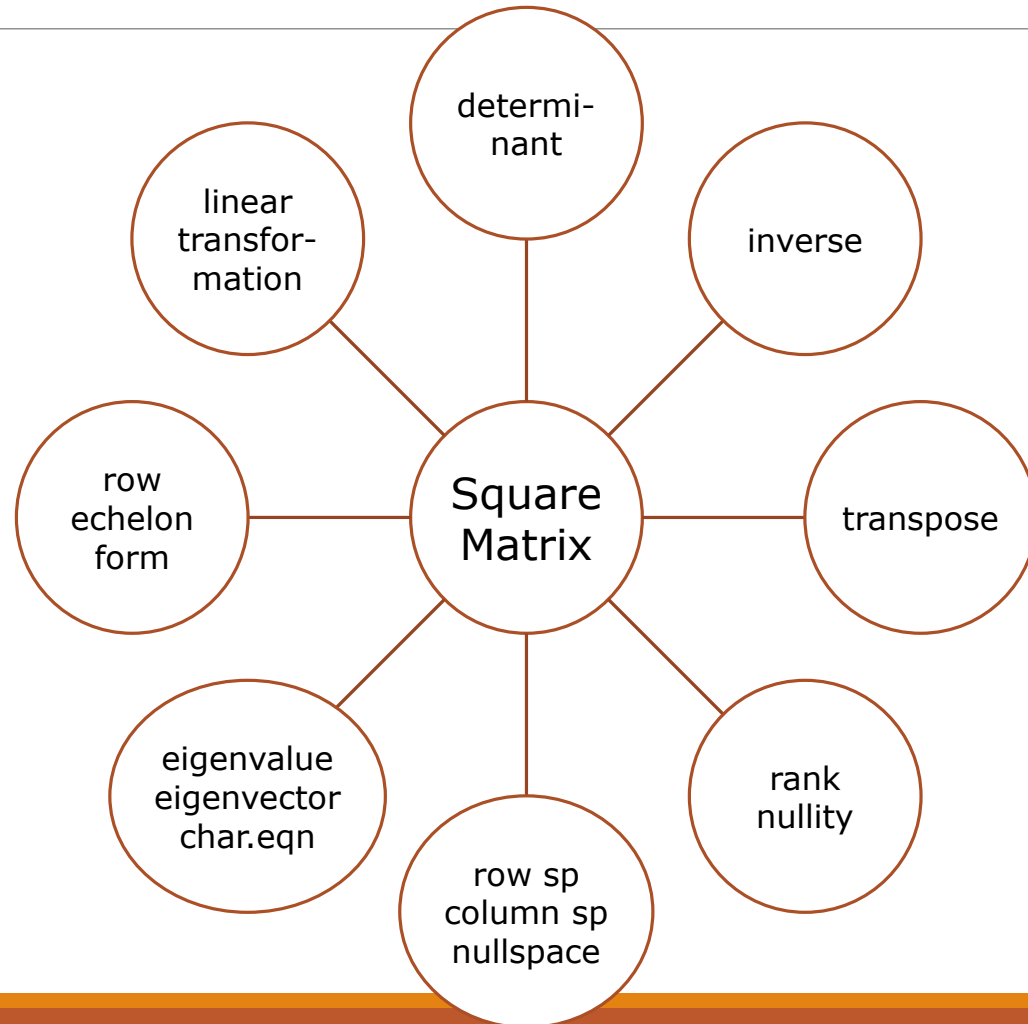
Linear Algebra with
Differential Equations

REVISION LECTURE

Main Concepts

Chapter	Tool kits (Main Concepts)	Applications (Linear model)
1	Linear system: row echelon form; Gaussian elimination; general solution Matrix: operations; inverse; non-singular; determinant; rank	$\mathbf{Ax} = \mathbf{b}$ (consistent) Data matrix
2	Vector space; n-space; linear combination; linear span; solution space; row/column/nullspace; linear independence; basis; dimension; coordinate vector; orthogonality; projection; least squares solutions	$\mathbf{Ax} = \mathbf{b}$ (inconsistent)
3	Linear transformation: standard matrix; geometrical transformation Diagonalization: characteristic polynomial; eigenvalue; eigenvector; eigenspace; powers of matrix	$\mathbf{x}_n = \mathbf{Ax}_{n-1}$
4	Homogeneous SDE: fundamental set of solution; general solution; equilibrium point; stability; trajectories; phase portraits	$\mathbf{y}' = \mathbf{Ay}$ IVP

Matrix & Associated Terminologies



Many faces of linear system

Standard form

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m \end{cases}$$

Augmented matrix form

$$\left(\begin{array}{cccc|c} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & b_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} & b_m \end{array} \right)$$

Vector equation form

$$x_1 \begin{pmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{pmatrix} + x_2 \begin{pmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{pmatrix} + \dots + x_n \begin{pmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}$$

Matrix equation form

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}$$

Solutions of linear system

no parameter
at least 1
parameter

$Ax = b$ Non-homogeneous	$Ax = 0$ Homogeneous
Maybe inconsistent	Always consistent
unique solution	only trivial solution
infinite solutions	trivial + non-trivial solutions
solution set	solution space
solutions not closed under addition and scalar mult.	solutions closed under addition and scalar mult.

$$\begin{array}{l} \text{general solution} \\ \text{of } Ax = b \end{array} = \begin{array}{l} \text{particular solution} \\ \text{of } Ax = b \end{array} + \begin{array}{l} \text{general solution} \\ \text{of } Ax = 0 \end{array}$$

Subset VS Subspace

- All **subspaces** of $\mathbf{R}^n$ are **subsets** of $\mathbf{R}^n$
- NOT all **subsets** of $\mathbf{R}^n$ are **subspaces** of $\mathbf{R}^n$
- A **subspace** of $\mathbf{R}^n$ is **closed** under addition and scalar multiplication
- Every **subspace** of $\mathbf{R}^n$ has a **basis** and a **dimension**
- A **subset** of $\mathbf{R}^n$ that is **not a subspace** does **not** have a **basis** or **dimension**
- If $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is a **subset** of $\mathbf{R}^n$,
then $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is a **subspace** of $\mathbf{R}^n$

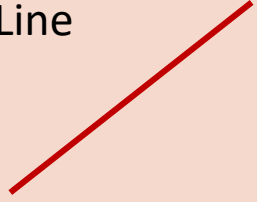
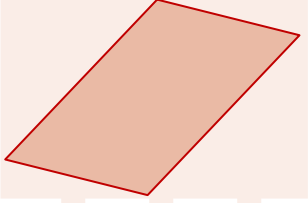
Some facts about basis for a vector space

- Basis is **not unique**.
- Different bases for same subspace have **same number of vectors**.
- The number of vectors in a basis is called the **dimension** of the subspace.
- A basis is the **largest set of linearly independent** vectors in the subspace.
- A basis is the **smallest set of vectors that can span the subspace**.
- Every vector can be written as a **linear combination** of the basis in a **unique way**.

Finding bases for some well known subspaces of $\mathbf{R}^n$

- **Span** $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ use column space method
- **Solution space** of $\mathbf{Ax} = \mathbf{0}$ use GE and separate parameters
- **Row space/ column space** of a matrix $\mathbf{A}$ use GE to get REF
- **nullspace** of a matrix $\mathbf{A}$ same as solution space of $\mathbf{Ax} = \mathbf{0}$
- **Eigenspace** associated with eigenvalue λ of a matrix $\mathbf{A}$
use GE to solve $(\lambda\mathbf{I} - \mathbf{A})\mathbf{x} = \mathbf{0}$
- A **line** in $\mathbf{R}^2$ and $\mathbf{R}^3$ that passes through the origin
just find a nonzero vector on the line
- A **plane** in $\mathbf{R}^3$ that contains the origin
just find two non-parallel vectors on the plane

Lines and Planes in $\mathbf{R}^3$

Object	Equation(s)	General solution	span
Line 	System: $ax + by + cz = d$ $ex + fy + gz = h$	$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = t \begin{pmatrix} p \\ q \\ r \end{pmatrix} + \begin{pmatrix} i \\ j \\ k \end{pmatrix}$	
Line through origin	Homog. system: $ax + by + cz = 0$ $ex + fy + gz = 0$	$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = t \begin{pmatrix} p \\ q \\ r \end{pmatrix}$	$\text{span}\{ \mathbf{v} \}$
Plane 	Linear equation: $ax + by + cz = d$	$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = t \begin{pmatrix} p \\ q \\ r \end{pmatrix} + s \begin{pmatrix} u \\ v \\ w \end{pmatrix} + \begin{pmatrix} i \\ j \\ k \end{pmatrix}$	
Plane through origin	Linear equation: $ax + by + cz = 0$	$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = t \begin{pmatrix} p \\ q \\ r \end{pmatrix} + s \begin{pmatrix} u \\ v \\ w \end{pmatrix}$	$\text{span}\{ \mathbf{u}, \mathbf{v} \}$

What's the use of G.E. (G.J.E.)?

1. Solve **linear system**
2. Find **inverse** of a matrix
3. Find **determinant** of a matrix
4. Find **linear combination** of a vector
5. Check **linear independence** of a set of vectors
6. Check whether a set of vectors **spans a subspace**
7. Find **coordinate vectors** (w.r.t. a basis)
8. Find **basis for row/column space** of a matrix
9. Find **rank** of a matrix
10. Find **eigenvector** / **generalized eigenvector** of a matrix
11. Find **formula** for a linear transformation (given images of a basis)
12. Find particular solution for IVP

Square matrix $\mathbf{A}$

$\mathbf{A}$ non-singular matrix

$\mathbf{A}$ has an inverse $\mathbf{A}^{-1}$

row echelon form of $\mathbf{A}$ does not have zero row

reduced row echelon form of $\mathbf{A}$ is the identity matrix

homogeneous system $\mathbf{Ax} = \mathbf{0}$
has only trivial solution

non-homogeneous system $\mathbf{Ax} = \mathbf{b}$
has exactly one solution

$\det \mathbf{A} \neq 0$

If $\mathbf{A}$ is $n \times n$, then $\text{rank } \mathbf{A} = n$

$\mathbf{A}$ is full rank

rows/columns of $\mathbf{A}$ are linearly independent

0 is not an eigenvalue of $\mathbf{A}$

$\mathbf{A}$ singular matrix

$\mathbf{A}$ does not have an inverse $\mathbf{A}^{-1}$

row echelon form of $\mathbf{A}$ has a zero row

reduced row echelon form of $\mathbf{A}$ is not the identity matrix

homogeneous system $\mathbf{Ax} = \mathbf{0}$
has both trivial and non-trivial solutions

non-homogeneous system $\mathbf{Ax} = \mathbf{b}$
has either no solution or infinitely many solutions

$\det \mathbf{A} = 0$

$\mathbf{A}$ is not full rank

If $\mathbf{A}$ is $n \times n$, then $\text{rank } \mathbf{A} < n$

rows/columns of $\mathbf{A}$ are linearly dependent

0 is an eigenvalue of $\mathbf{A}$